Matrix M (host)

$\boxed{\text{rowptr}_{\text{host}} = (1,\ 3,\ 4,\ 5,\ 6)}$

$\text{colval}_{\text{host}} = (2,\ 3,\ 1,\ 3,\ 4)$

$\text{nnzval}_{\text{host}} = (M_1,\ M_2,\ M_3,\ M_4,\ M_5)$

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} := \alpha \cdot \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & M_1 & M_2 & 0 \\ M_3 & 0 & 0 & 0 \\ 0 & 0 & M_4 & 0 \\ 0 & 0 & 0 & M_5 \end{bmatrix} \end{array} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \beta \cdot \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}$$

Distributed to devices

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} := \alpha \cdot \begin{array}{c} \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & M_1 & M_2 & 0 \\ M_3 & 0 & 0 & 0 \end{bmatrix} \end{array} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \beta \cdot \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

Matrix M, Tile 1 (device 1)

$\boxed{\text{rowptr}_1 = (1,\ 3,\ 4)}$

$\text{colval}_2 = (2,\ 3,\ 1)$

$\text{nnzval}_3 = (M_1,\ M_2,\ M_3)$

$$\begin{bmatrix} y_3 \\ y_4 \end{bmatrix} := \alpha \cdot \begin{array}{c} \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & 0 & M_4 & 0 \\ 0 & 0 & 0 & M_5 \end{bmatrix} \end{array} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \beta \cdot \begin{bmatrix} y_3 \\ y_4 \end{bmatrix}$$

Matrix M, Tile 2 (device 2)

$\boxed{\text{rowptr}_2 = (1,\ 2,\ 3)}$ ← offset by 4 against CPU replica

$\text{colval}_2 = (3,\ 4)$

$\text{nnzval}_2 = (M_4,\ M_5)$